3.2.4.2 Classification of Diaphragms

Diaphragms shall be classified as flexible when the maximum horizontal deformation of the diaphragm along its length is more than twice the average interstory drift of the vertical lateral-force-resisting elements of the story immediately below the diaphragm. For diaphragms supported by basement walls, the average interstory drift of the story above the diaphragm shall be used.

Diaphragms shall be classified as rigid when the maximum lateral deformation of the diaphragm is less than half the average interstory drift of the vertical lateral-force-resisting elements of the associated story.

Diaphragms that are neither flexible nor rigid shall be classified as stiff.

For the purpose of classifying diaphragms, interstory drift and diaphragm deformations shall be calculated using the pseudo lateral load specified in Equation (3-10). The in-plane deflection of the diaphragm shall be calculated for an in-plane distribution of lateral force consistent with the distribution of mass, and all in-plane lateral forces associated with offsets in the vertical seismic framing at that diaphragm level.

3.2.4.3 Mathematical Modeling

Mathematical modeling of buildings with rigid diaphragms shall account for the effects of horizontal torsion as specified in Section 3.2.2.2. Mathematical models of buildings with stiff or flexible diaphragms shall account for the effects of diaphragm flexibility by modeling the diaphragm as an element with an in-plane stiffness consistent with the structural characteristics of the diaphragm system. Alternatively, for buildings with flexible diaphragms at each floor level, each lateral-force-resisting element in a vertical plane shall be permitted to be designed independently, with seismic masses assigned on the basis of tributary area.

C3.2.4 Diaphragms

Evaluation of diaphragm demands should be based on the likely distribution of horizontal inertia forces. For flexible diaphragms, such a distribution may be given by Equation (C3-1) and illustrated in Figure C3-1 below.

$$f_d = \frac{1.5F_d}{L_d}\left[1 - \left(\frac{2x}{L_d}\right)^2\right] \qquad \text{(C3-1)}$$

where:

f_d = Inertial load per foot

F_d = Total inertial load on a flexible diaphragm

x = Distance from the center line of flexible diaphragm

L_d = Distance between lateral support points for diaphragm

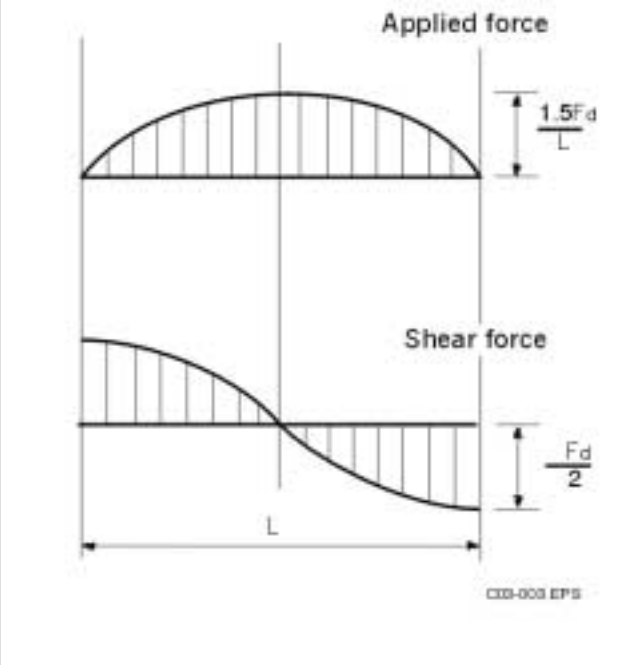

Figure C3-1 Plausible Force Distribution in a Flexible Diaphragm

3.2.5 P-Δ Effects

Buildings shall be evaluated for static P-Δ effects and dynamic P-Δ effects as specified in this section.

3.2.5.1 Static P-Δ Effects

Static P-Δ effects shall be included in linear or nonlinear analysis procedures as specified in Sections 3.2.5.1.1 and 3.2.5.1.2, respectively.